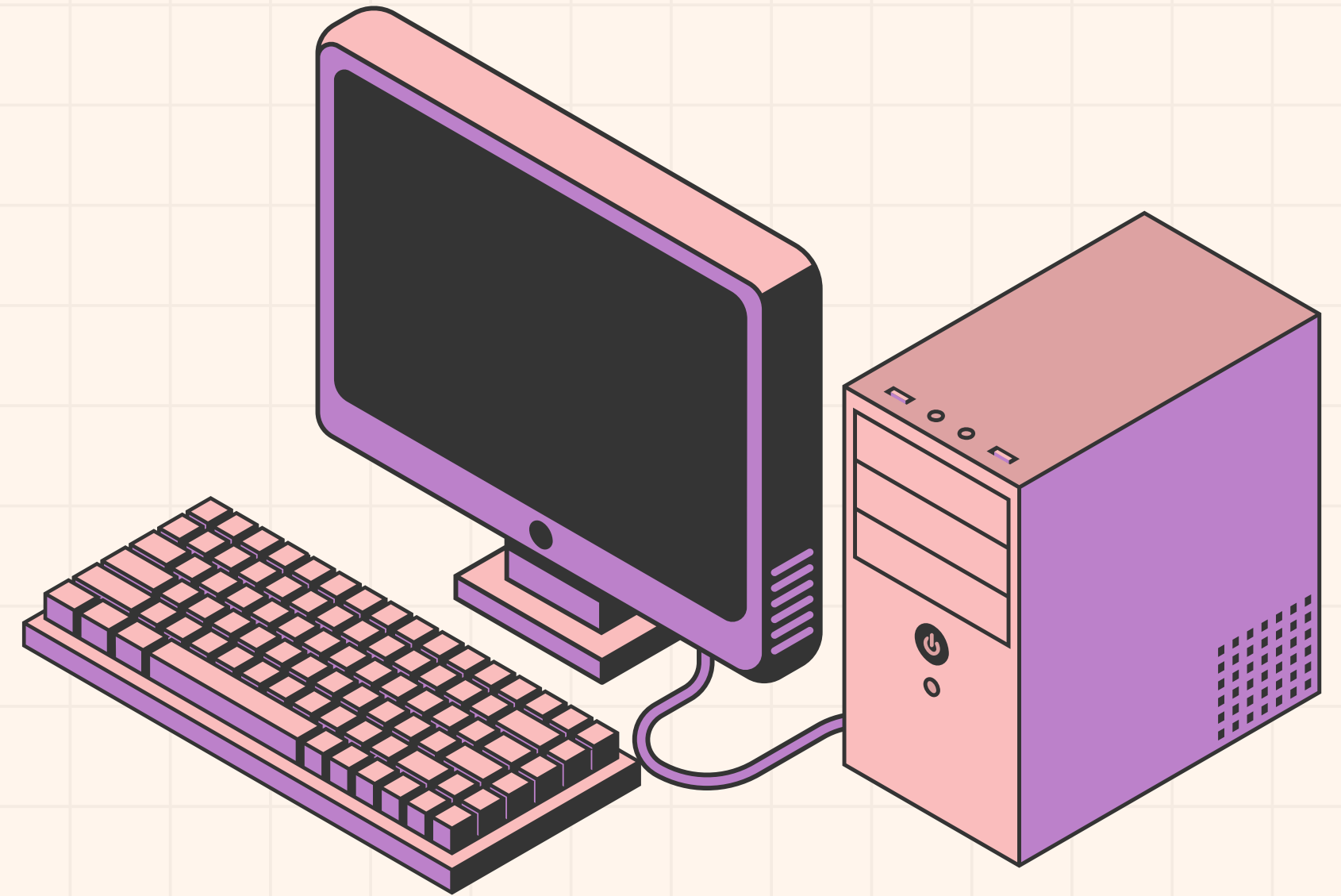


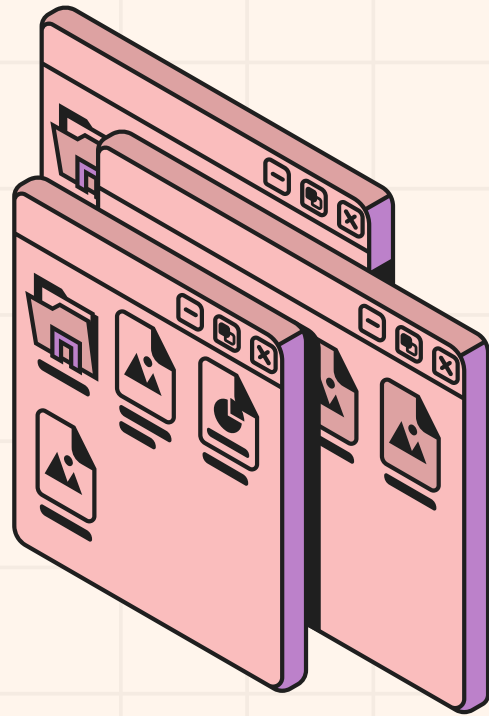
INTERNET

How it works



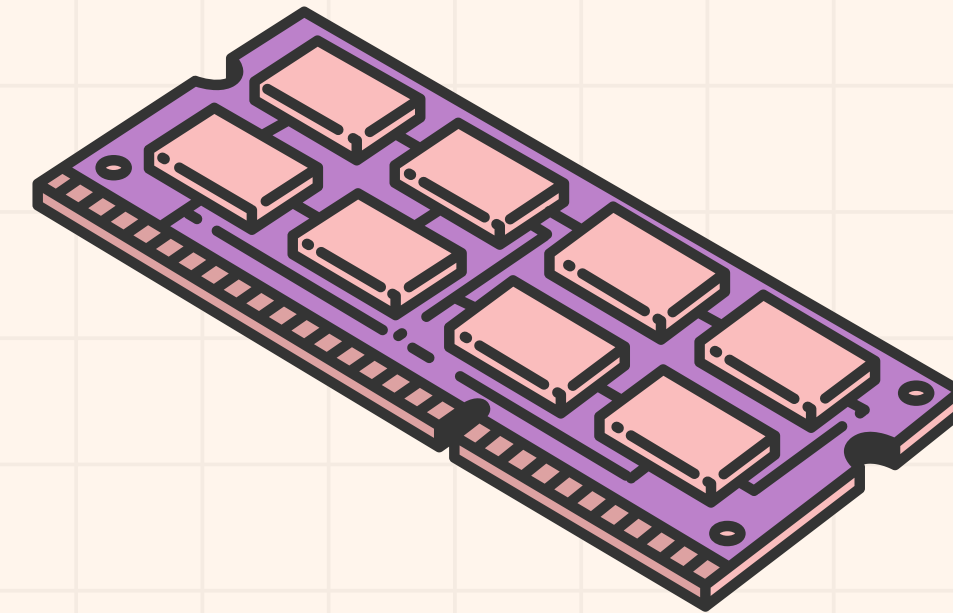
DATA TRANSFER

THE PROBLEM



COMPUTERS WERE ISLANDS

Unless a computer was directly plugged into a nearby computer, a computer could only have the data that was put on to it with a floppy disk. New research and important documents made on computers were isolated unless copied and mailed to a different computer.



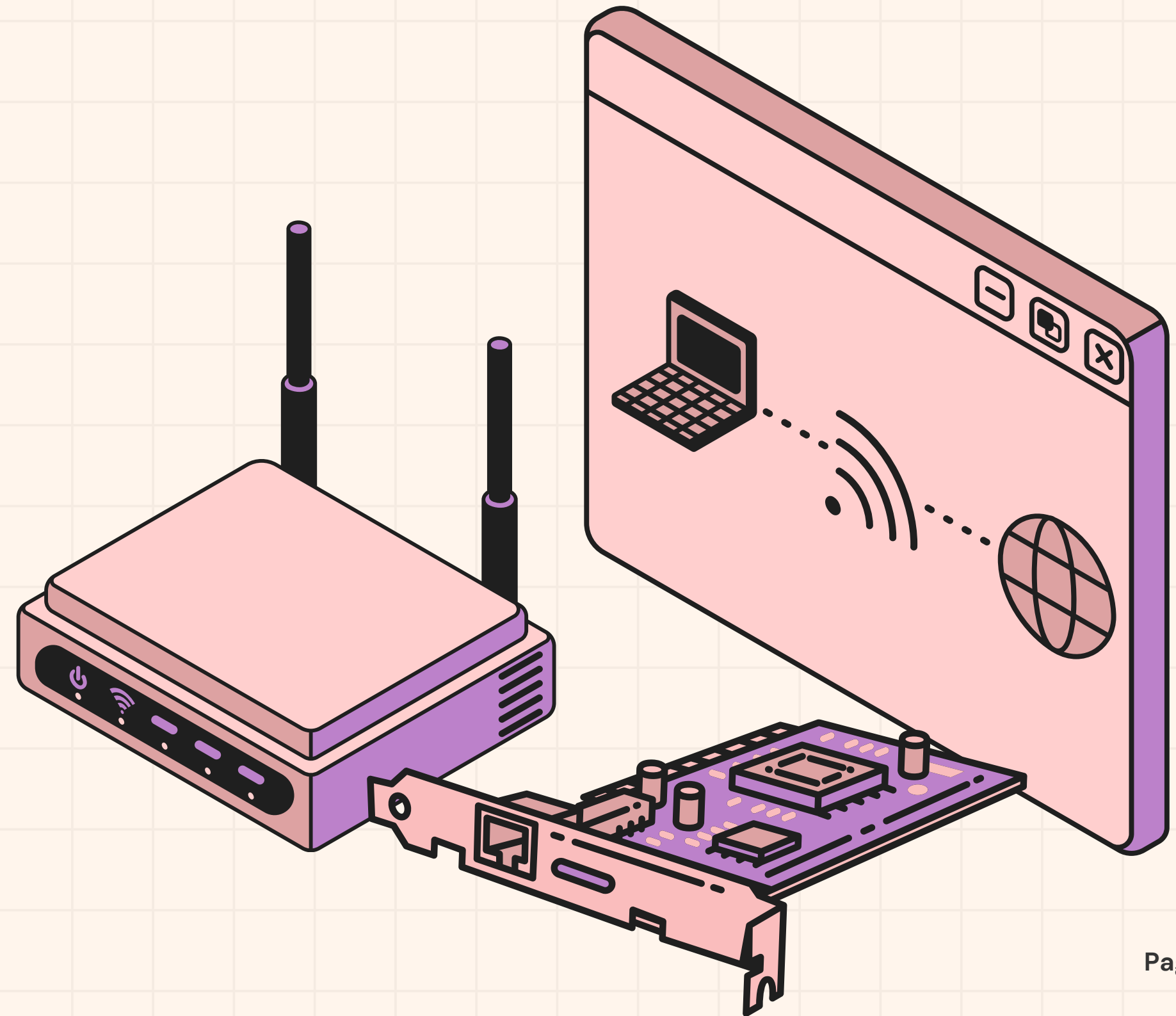
COLD WAR

Worries that the Soviet Union would blast the major centers of the US and Western Countries, being able to share data between knowledge centers became important. It was also important that computers were connected in multiple ways so that computers did not get stranded if the computers between them and others was taken out.

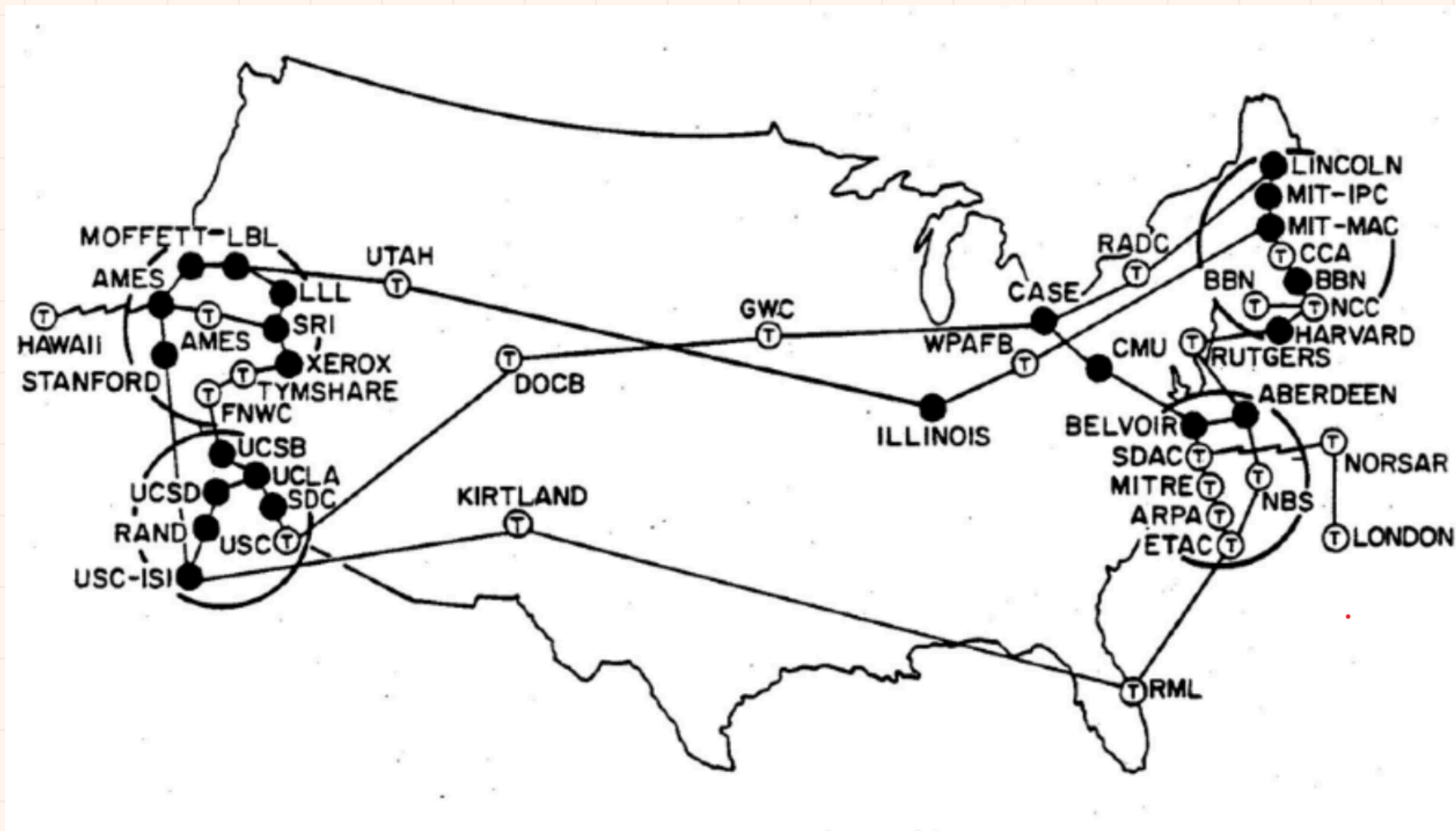
NETWORKING & THE INTERNET

A system using packets of data sent between computers was developed to solve these issues and allow data to be transferred. This system has been expanded today, but still uses the same protocols that were conceptualized in the 60s.

This system provided the robust, secure model that the US was looking for to connect the research computers around the country during the war.



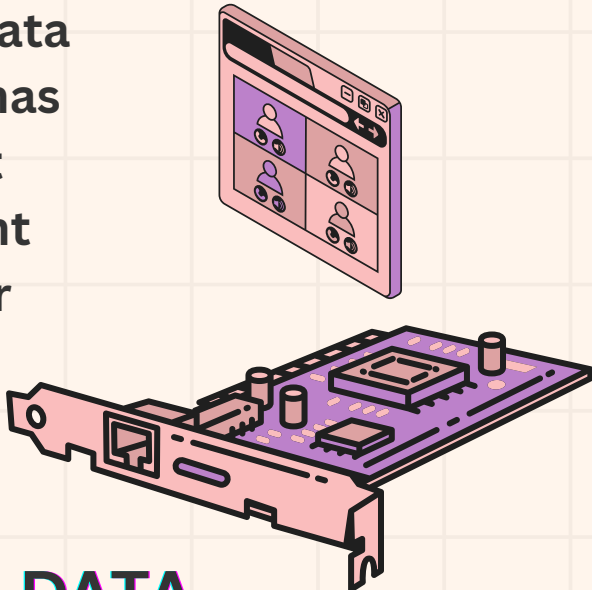
ARPANET



DATA TRAVEL

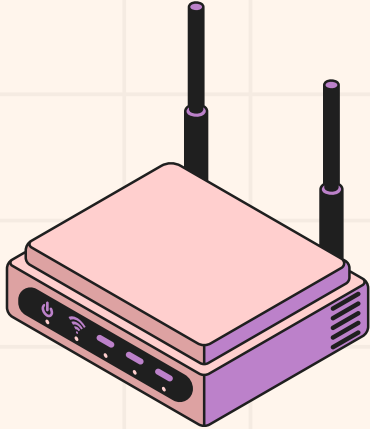
How Computers Send Data Over the Internet

This server has the data for a website a user has asked for. Data is put into a packet and sent to the server's router



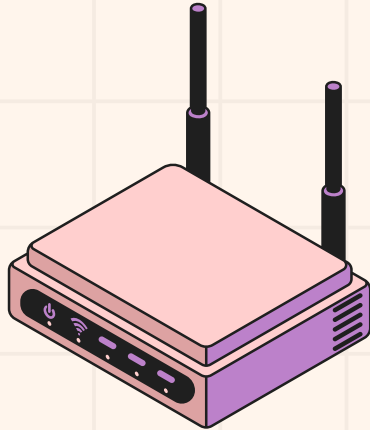
DATA

Sometimes the data passes through hundreds of routers, the routers work together to keep track of their connections so the destination in the header can be found



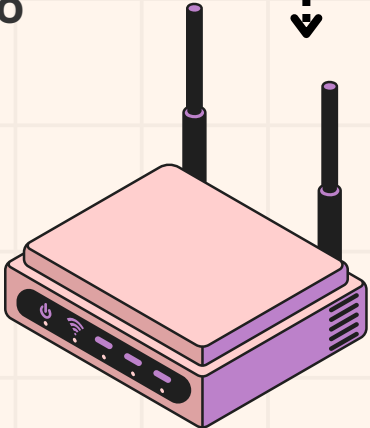
DATA | HEADER

The user's router receives the data, and takes off the header before passing it to the user's computer



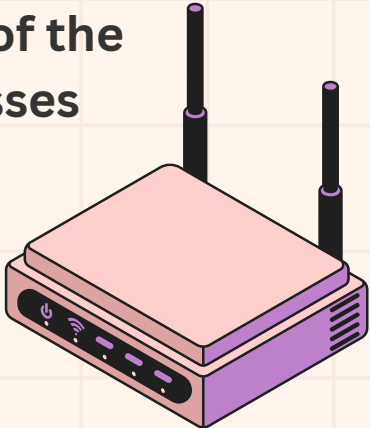
DATA | → HEADER

The router adds a frame header to the data, this has information of where the data needs to go, the router forwards the data



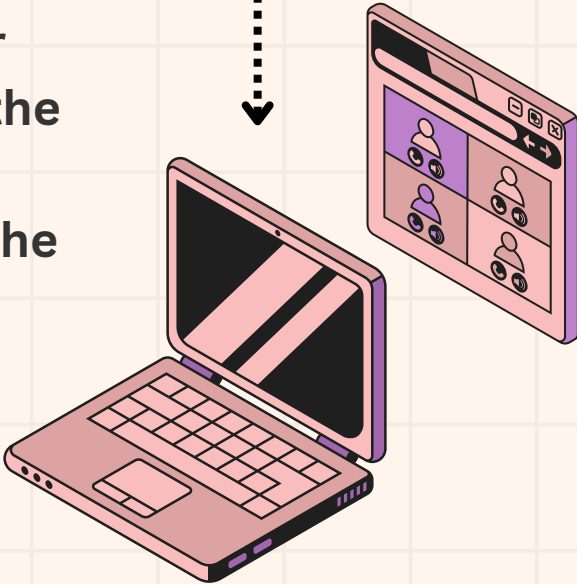
DATA | ← HEADER

The next router, without reading the data, uses the header to find the destination of the data and passes the data on



DATA | HEADER

The user's computer receives the data and displays the website



DATA

When a computer needs to send data to another computer on the internet, it sends the data and destination information to the router connecting the computer to the internet. This router puts the destination information into a header that is attached to the data. As the data is sent through the internet, no other router needs to read the data itself to ensure it gets to the right space.

The data is passed to other routers. These routers look to see if they are the destination, if not, they ask neighboring routers to pass the data onward. Once the destination of the data is found, the router connected to the destination takes the header off the data, and passes the data to the computer.

THANK YOU